

HETEROGENEOUS LLM FINE-TUNING



CENTRAL SERVER (SPA Aggregation)

1. Reconstruct

Input: Ranks
 $\Delta W = B \times A$



2. SVD

Extract Signal
 $W = U \Sigma V^T$

3. Project

Adaptive Rank
 $r=4, r=16...$



RESULTS (Yelp)

✓ **Acc: 63.8%** (vs 61%)
✓ **Cost: -25%**
✓ **PPL: 11.94**



Client A (Edge)

8GB | $r=4$



Frozen LLM



⚙️ LoRA (Small)

$B_4 \times A_4$



Client B (Mid)

16GB | $r=8$



Frozen LLM



⚙️ LoRA (Med)

$B_8 \times A_8$



Client C (High)

24GB | $r=16$



Frozen LLM



⚙️ LoRA (High)

$B_{16} \times A_{16}$



Client D (Server)

48GB | $r=32$



Frozen LLM



⚙️ LoRA (Large)

$B_{32} \times A_{32}$

HETEROGENEOUS CLIENT LAYER (Federated)